

Technical Guide

NeXtTimber®
by Timberlink

IT'S WHAT BETTER TOMORROWS ARE BUILT ON

Version 3
April 2026

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Introduction



Overview

A Link from Seed to Structure

NeXTimber® by Timberlink manufactures Australian made Engineered Wood Products (EWP) and provides mass timber building solutions with Cross Laminated Timber (CLT) panels and Glue Laminated Timber (GLT) members.

Made from responsibly sourced Australian plantation pine, NeXTimber® CLT panels and GLT members offer long term carbon capture and storage like traditional timber but with the strength, uniformity and technical predictability to support increasingly demanding applications such as hybrid and tall wood buildings.

Whilst NeXTimber® does not offer structural engineering design services or certification, our locally based technical and support team is available to provide support to help optimise the use of NeXTimber® products in your project and help to achieve cost, time and environmental efficiencies.

NeXTimber® CLT panels and GLT members are produced at a purpose-built manufacturing plant located adjacent to Timberlink's timber processing facility in Tarpeena, South Australia. NeXTimber® EWPs are manufactured from Timberlink timber, responsibly sourced from Australian pine plantations.

NeXTimber® can offer support in technical, logistical and planning for project teams utilising NeXTimber® products to suit all project designs, procurement and delivery models. The level of support can be tailored to the client's and project team's requirements.

Responsibility	Client Team	NeXTimber®
Building Design and Documentation	●	Advisory Support
Connection Details and Specification	●	Advisory Support
Structural Panelisation	●	Advisory Support
Engineering Design and Certification	●	Advisory Support
Refined Panelisation for Manufacture		●
Shop Drawings		●
Take Offs		●
Supply of CLT, GLT & Connections		●
Installation/Assembly	●	Advisory Support

Typical roles and responsibilities for a project with NeXTimber® CLT and GLT

Overview continued

Document Scope

This information and advice in this Technical Guide is of a general nature only and has not been prepared with your specific project needs in mind. You should always obtain specialist advice to ensure that the products referred to in this Technical Guide meet your specific requirements.

This Technical Guide and information is specific to the NeXTimber® products referred to within it and no warranty is given to the suitability and application of the information to other manufacturers of similar products.

This Technical Guide provides key information necessary for the pre-analysis assessment of NeXTimber® products. It covers material properties, design principles, connections and other general performance requirements, and includes pre-analysis product selection tables derived from product testing and analysis.

Application of this Technical Guide by Suitable Professionals

It is the responsibility of the user to ensure that the use of this Technical Guide and NeXTimber® products is appropriate to the relevant application and to exercise their own professional judgement in doing so. This guide has been prepared for use by suitably qualified construction professionals to assist in the design and specification of NeXTimber® products. Due regard should be given by the suitably qualified professional when using this Technical Guide to the relevant Standards and all other relevant codes and industry best practice guidelines when assessing the use of a NeXTimber® product. Advice on overall building design issues are not covered by this guide and advice should be sought from suitably qualified professionals when using NeXTimber® products as part of an overall building system.

Full responsibility for the design and general compliance with all laws including the Building Codes of Australia, National Construction Code of Australia (NCC) and New Zealand Building Code (NZBC) and all relevant Standards, codes and guidelines rests with the design professional specifying and certifying the use and application of the product. NeXTimber® will not accept any liability for the failure of any other elements of a building which cause a subsequent failure of an NeXTimber® product.

NeXTimber® Technical Information

This Technical Guide is current as at the date of issue. It is the user's responsibility to ensure that the latest version of this Technical Guide is in use at all times. Please check the NeXTimber® website for the latest version of this Technical Guide and other NeXTimber® documentation including specific guides and test information.

Access all the latest information and downloads from the NeXTimber® website: [NeXTimber.com.au](https://www.nexttimber.com.au)

Mass Timber

Broadly speaking, mass timber comprises of engineered wood products that are laminated and pressed together to form large timber elements that are strong, stable and capable of withstanding large loads.

These elements can be used to build a multitude of buildings; from residential homes to large multi-storey open plan spaces. More importantly, mass timber is a prefabricated building system that utilises automated manufacturing techniques that allow it to be fast, safe, and sustainable whilst benefiting offsite construction.



Photos of NeXTimber CLT & GLT manufactured with Timberlink Radiata pine

What is NeXTimber® CLT?

NeXTimber® Cross Laminated Timber is made from machine graded, kiln-dried softwood, which is finger jointed, dressed and arranged in multiple layers (lamellas) of 3, 5, 7 or 9. Alternating layers are laid perpendicular to each other, and each layer is glued along the face and edges of each lamella before being cured under pressure to form one solid rectangular panel (or billet). Individual building elements are then digitally placed and cut from the billet using Computer Numerical Controlled (CNC) technology which includes a combination of saws, routers and mills. The layup of the panels, combined with the digital fabrication, offer a strong, rigid and dimensionally stable and highly accurate building product that offers versatility for designers.

The cross-laminated nature of the panels creates a product that offers rigidity in two directions. CLT is a panelised system and is well suited for use as subfloors, walls, floors, shafts and roof elements.

What is NeXTimber® GLT?

NeXTimber® Glue Laminated Timber is an engineered wood product made from finger-jointed and laminated layers of machine graded, kiln-dried and dressed timber, pressed together with an adhesive under pressure to form one singular element, which creates a high performing structural member.

Unlike CLT, the grain runs only in the primary direction of the element, making GLT well suited for use as beams, columns and portal frame structures.

NeXTimber® Features and Benefits

NeXTimber® provides a unique opportunity for the building industry to utilise high performing, locally manufactured engineered wood products.

Features and Benefits of NeXTimber® CLT & GLT



Premium Product Performance

- ✓ Made from consistent high grade feedstock, structurally graded to AS/ NZS 1748.1 and verified to AS/NZS 4490 from Timberlink's Tarpeena manufacturing facility.
- ✓ NeXTimber® CLT panels are edge glued and sanded to provide a clean, calibrated panel surface finish with no gaps between the lamellas.
- ✓ The melamine adhesive used in NeXTimber® CLT and GLT products is resistant to heat and the products do not exhibit premature heat-induced char fall-off as demonstrated through full-scale furnace tests, carried out in accordance with AS 1530.4.
- ✓ QA processes from infeed to installation and certified manufactured to ISO 16696-1.



Local Australian Supply

- ✓ All infeed stock is responsibly sourced from local Australian pine plantations.
- ✓ One supply point for both CLT and GLT with associated fixings.
- ✓ Local Australian supply can provide more flexibility in lead times to give design and build teams more opportunities for optimisation.
- ✓ Reduced risk to project budgets & timelines from international shipping, import taxes and currency exchange risks.
- ✓ Section sizes aren't restricted by shipping container limits which allows for flexibility in design and more efficient installation.
- ✓ Locally based technical support throughout a project's lifecycle.



The Ultimate Renewable™

- ✓ NeXTimber® feedstock sourced from co-located Timberlink mill has dual certified sustainability credentials.
- ✓ NeXTimber® products are FSC® *chain of custody certified (FSC C117015)* and Programme for the Endorsement of Forest Certification™ (PEFC) / Responsible Wood (RW) certified. By choosing Timberlink® Australia products like NeXTimber®, you are supporting the responsible management of the world's forests.
- ✓ Timber is the Ultimate Renewable™. NeXTimber® products provide a prime opportunity to reduce the embodied carbon of a project.

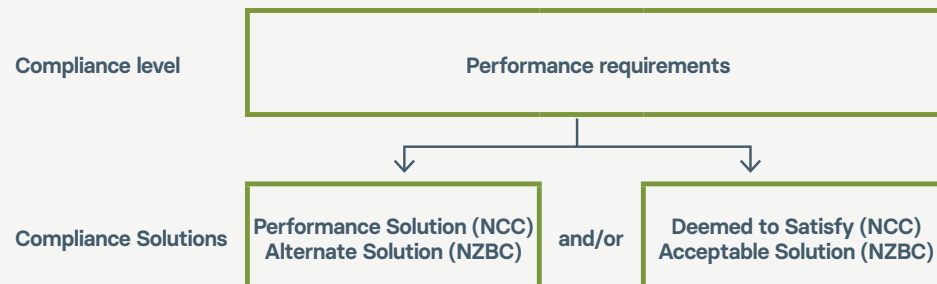
A modern office interior with wood-paneled walls and ceiling. Exposed metal pipes run along the ceiling, with several black industrial-style pendant lights hanging from them. On the left, a bicycle is mounted on the wall. In the center, a long wooden desk is set up with multiple computer monitors and ergonomic chairs. Large windows in the background let in natural light. On the right, a large potted plant with broad green leaves sits in a grey metal container. A blue diagonal graphic overlay covers the bottom-left corner of the image.

NCC and NZBC Compliance

National Construction Code (NCC) and New Zealand Building Code Compliance (NZBC)

Overview

All building structures are required to meet the National Construction Code of Australia (NCC) performance requirements to ensure the building meets a minimum level of performance for the health and safety of its occupants. The NCC's Performance Requirements can be met using either a Performance Solution (Alternative Solution) or a Deemed-to-Satisfy (DTS) Solution, or a combination of both.



Compliance method to meet the NCC's Performance Requirements

NCC Compliance - CLT

In accordance with NCC referenced Standard AS 1720.1 NeXTimber® CLT complies with AS 1720.1, through Clause 1.3.1 (h) Other timber products and grades, as its characteristic properties, duration of load effects, and structural durability of adhesives, have been established by test or report. However, at present AS 1720.1 – Design of Timber Structures, does not explicitly cover the design procedure for CLT. Design professionals will need to determine section properties based on industry accepted methods and apply the appropriate strength and serviceability structural design checks to meet compliance with the NCC.

GLT Compliance

NeXTimber® GLT is manufactured to meet Australian Standard, AS 1328.1 – Glue Laminated Structural Timber and may be designed in accordance with AS 1720.1 – Design of Timber Structures, Part 1: Design Methods.

NZBC Compliance

CLT Cross Laminated Timber design is not referenced in the New Zealand Building Code (NZBC), and is not explicitly covered in the design standards including AS 1720.1 Timber Structures: part 1 Design Methods and NZS 3603:1993 Timber Structures Standard. To comply with the NZBC, designers need to demonstrate structural stability and durability is achieved for the required design life, and engineers must design and certify the design as part of an Alternative Solution pathway. Evidence is dependent on the consultant carrying out the design and can be demonstrated by adopting design procedures outlined in the following publications, namely (but not limited to):

- FP-Innovation CLT Handbook - 2019
- ProHolz Cross-Laminated Timber Structural Design Volume 1 : 2014
- ProHolz Cross-Laminated Timber Structural Design Volume 2 : 2018
- Swedish Wood - The CLT Handbook – 2019 The principles of strength and serviceability checks used by designers generally follow the provisions laid out for the design of timber in AS 1720.1 and/or NZS 3603:1993.





NeXTimber® Specifications

CLT Range

Technical Specification

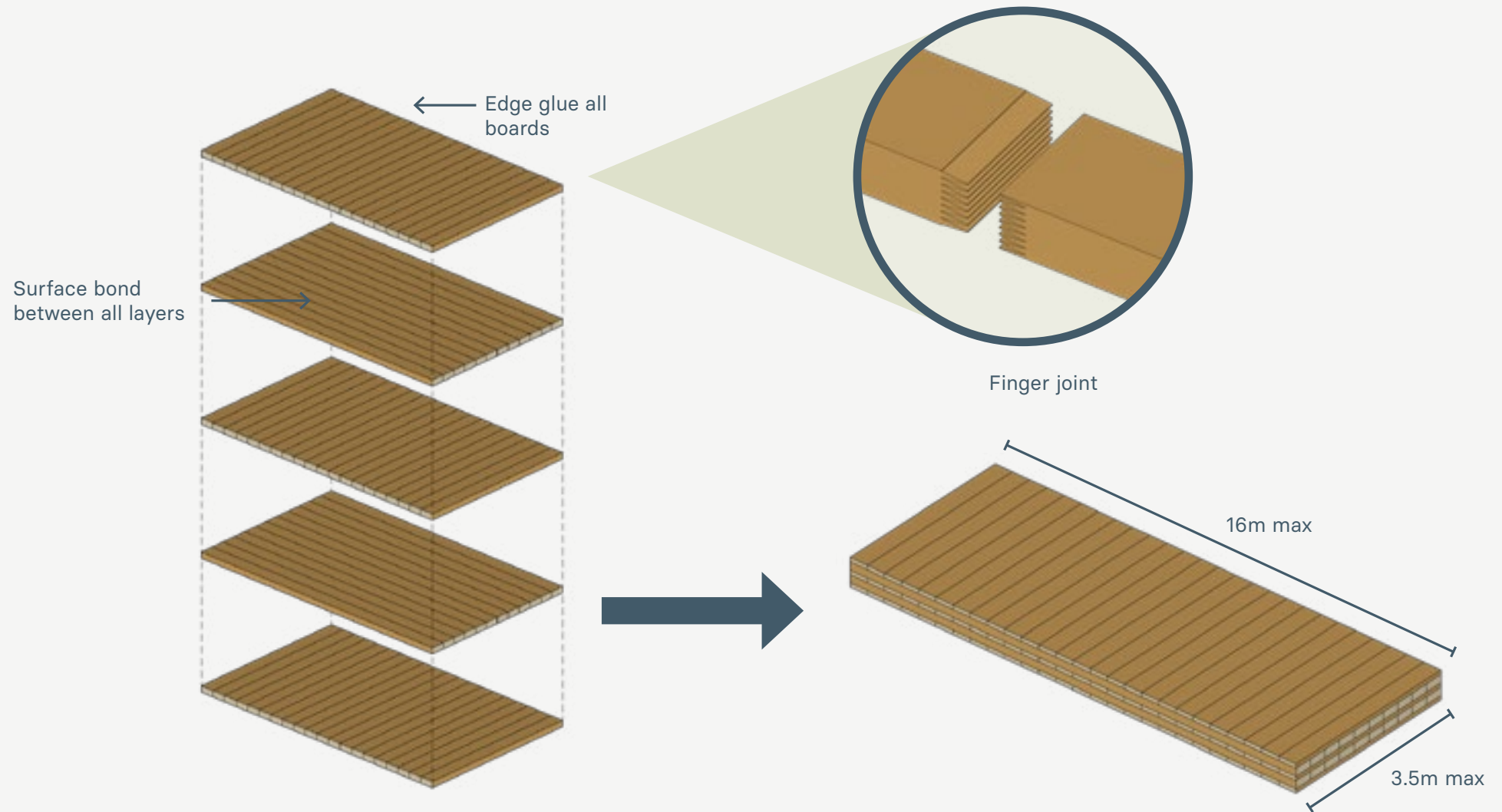
Wood Species	Australian Grown Radiata Pine (Pinus Radiata) Kiln Dried Softwood
Mechanical Grading	In accordance with AS/NZS 1748.1 and AS/NZS 4490
Lamella	20, 30 and 40 mm
Lamella Stress Grade	MGP10
Joint Group	JD5
Strength Group	SD6
Moisture Content ex. Factory	10-15%
Maximum Panel Dimensions	3.5 m wide x 16m long
Laminations	3 to 9
Panel Thickness	90 mm to 360 mm
Feedstock Perseverative Treatments	Untreated or preservative treated ³ to Hazard Class 3 in accordance with AS/NZS 1604.1
Adhesive – Face and edge	Melamine Formaldehyde
Adhesive – Finger Joints	Melamine Formaldehyde
Manufacturing Standards	ISO 16696-1

Technical Specification

Surface Quality	Natural grade, sanded finish. Edge-glued with mechanically calibrated surface (minimum 60 grit sandpaper). Sanding direction is perpendicular to grain on crosswise layups.
Density	460 to 540 kg/m ³
Fire Hazard Properties – Group Number and Average specific extinction area¹	Radiata pine: Group 3 and < 10 m ² /kg
Fire Hazard Properties – Critical Radiant Flux and Smoke Development rate²	Radiata pine: 3.5 kW/m ² and 7 % – min
Spread of Flame and Smoke Development Index	Radiata pine: 8 and 3
Thermal Conductivity	U = 0.12 W/mK at 12% MC (R=t/U)

1. An Assessment of Solid Timber Wall and Ceiling Linings in Accordance with AS 5637.1:2015 Report Number: RIR 45980.10
2. Fire Hazard Properties of Timber Floor, Wall and Ceiling Linings in Accordance with the Requirements of NCC 2019 Volume One Report Number: RIR 41117.9
3. Although treated, finished product is not warranted for use in external, weather exposed applications.

CLT Range continued

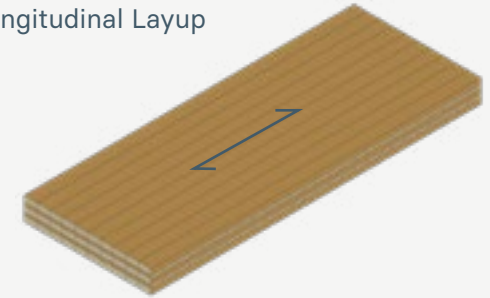


CLT Range continued

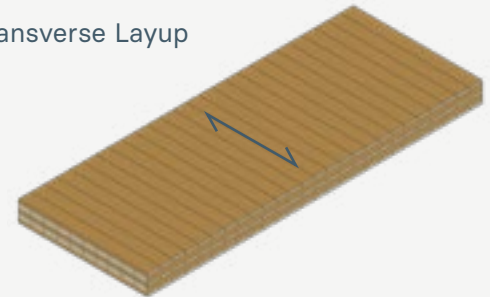
Panel Layups

Panel Designation	Thickness	Layers	Lamella Thickness (mm)									
NX 3-90	90	3	30	30	30							
NX 3-100	100	3	30	40	30							
NX 3-110	110	3	40	30	40							
NX 3-120	120	3	40	40	40							
NX 5-130	130	5	30	20	30	20	30					
NX 5-140	140	5	40	20	20	20	40					
NX 5-150	150	5	30	30	30	30	30					
NX 5-160	160	5	30	30	40	30	30					
NX 5-170	170	5	40	30	30	30	40					
NX 5-180	180	5	40	30	40	30	40					
NX 5-190	190	5	40	40	30	40	40					
NX 5-200	200	5	40	40	40	40	40					
NX 7-210	210	7	30	30	30	30	30	30	30			
NX 7-220	220	7	30	30	30	40	30	30	30			
NX 7-230	230	7	40	30	30	30	30	30	30	40		
NX 7-240	240	7	40	30	30	40	30	30	30	40		
NX 7-250	250	7	40	30	40	30	40	30	40	30	40	
NX 7-260	260	7	40	30	40	40	40	40	30	40		
NX 7-270	270	7	40	40	40	30	40	40	40	40		
NX 7-280	280	7	40	40	40	40	40	40	40	40		
NX 9-290	290	9	40	30	30	30	30	30	30	30	30	40
NX 9-300	300	9	40	30	30	30	40	30	30	30	30	40
NX 9-310	310	9	40	30	30	40	30	40	30	30	30	40
NX 9-320	320	9	40	30	30	40	40	40	30	30	30	40
NX 9-330	330	9	40	30	40	40	30	40	40	40	30	40
NX 9-340	340	9	40	30	40	40	40	40	40	40	30	40
NX 9-350	350	9	40	40	40	40	30	40	40	40	40	40
NX 9-360	360	9	40	40	40	40	40	40	40	40	40	40

Longitudinal Layup



Transverse Layup



Note: Standard Dimension

Width: 2.4 to 3.5m in 0.1m increments

Length: 6m to 16m in 0.1m increments

Treatments: Untreated and H3 Treated

Nomenclature: Panel Designation - Treatment

i.e NX3-120 H3

Pre-Analysis Information

The information within this technical guide outlines the structural properties of our engineered wood products range. This document contains pre-analysis tables that provide indicative spanning capabilities of NeXTimber® panels based on standard layups.

The tables are prepared to inform the designer of indicative panel sizes and inform suitable panel size selection prior to commencing the detailed design of a project or used to develop project budgets. Designers need to consider penetrations, voids, stiffness of supporting elements and project-specific acceptance criteria that may influence the final sizing of the panels.

Pre-analysis table notes:

- Self weight of panels is included in the assessment. Additional dead loads shall be considered as part of the Superimposed Dead Load
- Analysis assumes uniformly distributed loads across all spans. Pattern loading has not been considered
- Analysis assumes the panels are supported on rigid supports. Floor vibrations and compound deflections need to be assessed separately
- Long term deflections factor, $j_2 = 2$
- Capacity factor, $\phi = 0.85$
- Partial seasoning factor, $k_4 = 1$
- Temperature effects, $k_6 = 1$

The information within the pre-analysis tables provided is not structurally certified. The information is for pre-analysis purposes only. The project structural engineer must undertake a full detailed analysis of all NeXTimber® CLT and GLT elements as a part of the certification of the structural system to provide evidence of compliance with the NCC or NZBC as applicable.

Feedstock Mechanical Properties - Cross Laminated Timber

Modulus of Elasticity	Parallel	10,000 MPa
	Perpendicular	333 MPa
Shear Modulus, G_0	Parallel	667 Mpa
Rolling Shear Modulus, G_r	Perpendicular	67 MPa
Bending Strength	Parallel	17 MPa
Tensile Strength	Parrallel	7.7 MPa
	Perpendicular	0.5 MPa
Compressive Strength	Parrallel	18 MPa
	Perpendicular	10 MPa
Shear Strength	Parrallel	2.6 MPa
Rolling Shear Strength	Perpendicular	1.2 Mpa

Note: Charateristic values supplied above are for feedstock properties conforming with AS 1720.1 Table H3.1. ProHolz design procedure recommends limiting perpendicular compressive stress to 3MPa.

CLT Pre-Analysis Tables

Floor Pre-Analysis Tables

The tables within this CLT section have been prepared using the Shear Analogy Method, in line with the procedure suggested in FPIInnovations: CLT Handbook, 2019 Edition and WoodSolutions Design Guide 50. The performance of the panels is based on empirical methods and takes into account the shear deformation of the transverse boards. These tables are for ambient conditions, and a separate assessment needs to be undertaken for the AS 1170.1 fire load case related to the project's Fire Resistance Level (FRL), discussed in the Fire Resistance Level section of this guide.

The recommended spans of the panels within the tables have been prepared on the basis of the following acceptance criteria. The designer may wish to increase or decrease these criteria based on the project demands. If comparing products between different suppliers, it is recommended that designers review the actual panel stiffness and structural performance as the criteria used for the development of span tables differs from supplier to supplier and may not reflect the project's performance criteria.

Permanent Load (Dead Load) (including superimposed dead load), G	Span/250
Imposed Loads (Live Load), Q	Span/300
Long Term, $j2 \times (G + 0.4Q)$	The lesser of, Span/300 and 25 mm

Floor Vibration

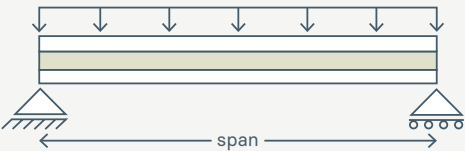
Due to the relatively low density of mass timber, dynamic performance is an important design consideration. Typically, for longer spanning floors, the floor performance will be governed by the perceived dynamic performance of the element. This design consideration is particularly important in residential buildings or where there may be sensitive equipment. The dynamic response of horizontal spanning elements is influenced by:

- Rigidity of supporting elements
- Mass of the structure
- Damping

The figures published in this technical guide are based on the following performance criteria:

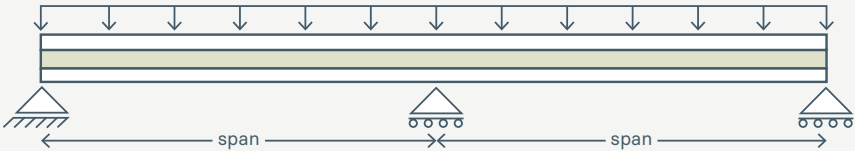
Vibration	Minimum of 8 Hz or where the natural frequency is less than 8 Hz, the acceleration does not exceed 0.05 m/s ²
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Floor – Pre-Analysis Tables



Single span

Live Load (kPa)	2			3			5		
Superimposed Dead Load (kPa)	0.5	1	2	0.5	1	2	0.5	1	2
Span (mm)									
2500	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-100
2750	NX3-90	NX3-90	NX3-100	NX3-90	NX3-90	NX3-100	NX3-100	NX3-100	NX3-110
3000	NX3-90	NX3-90	NX3-100	NX3-100	NX3-100	NX3-110	NX3-110	NX3-110	NX3-120
3250	NX3-100	NX3-100	NX3-110	NX3-100	NX3-110	NX3-120	NX3-120	NX3-120	NX5-140
3500	NX3-100	NX3-110	NX3-120	NX3-110	NX3-120	NX5-130	NX5-130	NX5-140	NX5-140
3750	NX3-110	NX3-120	NX5-140	NX3-120	NX5-130	NX5-140	NX5-140	NX5-140	NX5-150
4000	NX3-120	NX5-130	NX5-140	NX5-130	NX5-140	NX5-150	NX5-150	NX5-150	NX5-160
4250	NX5-130	NX5-140	NX5-150	NX5-140	NX5-150	NX5-160	NX5-160	NX5-160	NX5-180
4500	NX5-140	NX5-150	NX5-160	NX5-150	NX5-160	NX5-170	NX5-170	NX5-180	NX5-190
4750	NX5-140	NX5-150	NX5-180	NX5-160	NX5-170	NX5-190	NX5-180	NX5-190	NX7-210
5000	NX5-150	NX5-160	NX5-200	NX5-160	NX5-180	NX5-200	NX5-190	NX5-200	NX7-220
5250	NX5-160	NX5-180	NX7-220	NX5-180	NX5-190	NX7-220	NX5-200	NX7-220	NX7-230
5500	NX5-170	NX5-190	NX7-230	NX5-190	NX5-200	NX7-230	NX7-220	NX7-220	NX7-240
5750	NX5-180	NX7-220	NX7-250	NX5-200	NX7-220	NX7-250	NX7-220	NX7-230	NX7-250
6000	NX5-200	NX7-220	NX7-260	NX7-220	NX7-220	NX7-260	NX7-240	NX7-250	NX7-260
6250	NX7-220	NX7-240	NX9-290	NX7-220	NX7-240	NX9-290	NX7-250	NX7-260	NX9-290
6500	NX7-230	NX7-260	NX9-300	NX7-230	NX7-260	NX9-300	NX7-260	NX7-280	NX9-300
6750	NX7-250	NX7-280	NX9-320	NX7-250	NX7-280	NX9-320	NX7-280	NX9-290	NX9-320
7000	NX7-260	NX9-300	NX9-340	NX7-260	NX9-300	NX9-340	NX9-290	NX9-300	NX9-340
7250	NX9-290	NX9-320		NX9-290	NX9-320		NX9-300	NX9-320	
7500	NX9-300	NX9-330		NX9-300	NX9-330		NX9-310	NX9-330	
7750	NX9-330	NX9-360		NX9-330	NX9-360		NX9-330	NX9-360	
8000	NX9-340			NX9-340			NX9-340		



Double span

Live Load (kPa)	2			3			5		
Superimposed Dead Load (kPa)	0.5	1	2	0.5	1	2	0.5	1	2
Span (mm)									
2500	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90
2750	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90
3000	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90
3250	NX3-90	NX3-90	NX3-100	NX3-90	NX3-90	NX3-100	NX3-90	NX3-90	NX3-100
3500	NX3-90	NX3-90	NX3-110	NX3-90	NX3-90	NX3-110	NX3-90	NX3-100	NX3-110
3750	NX3-90	NX3-100	NX3-120	NX3-90	NX3-100	NX3-120	NX3-100	NX3-100	NX3-120
4000	NX3-100	NX3-110	NX5-140	NX3-100	NX3-110	NX5-140	NX3-110	NX3-110	NX5-140
4250	NX3-110	NX3-120	NX5-150	NX3-110	NX3-120	NX5-150	NX3-110	NX3-120	NX5-150
4500	NX3-120	NX5-140	NX5-160	NX3-120	NX5-140	NX5-160	NX3-120	NX5-140	NX5-160
4750	NX5-130	NX5-150	NX5-180	NX5-130	NX5-150	NX5-180	NX5-130	NX5-150	NX5-180
5000	NX5-140	NX5-160	NX5-200	NX5-140	NX5-160	NX5-200	NX5-140	NX5-160	NX5-200
5250	NX5-150	NX5-180	NX7-220	NX5-150	NX5-180	NX7-220	NX5-150	NX5-180	NX7-220
5500	NX5-170	NX5-190	NX7-230	NX5-170	NX5-190	NX7-230	NX5-170	NX5-190	NX7-230
5750	NX5-180	NX7-220	NX7-250	NX5-180	NX7-220	NX7-250	NX5-180	NX7-220	NX7-250
6000	NX5-200	NX7-220	NX7-260	NX5-200	NX7-220	NX7-260	NX5-200	NX7-220	NX7-260
6250	NX7-220	NX7-240	NX9-290	NX7-220	NX7-240	NX9-290	NX7-220	NX7-240	NX9-290
6500	NX7-230	NX7-260	NX9-300	NX7-230	NX7-260	NX9-300	NX7-230	NX7-260	NX9-300
6750	NX7-250	NX7-280	NX9-320	NX7-250	NX7-280	NX9-320	NX7-250	NX7-280	NX9-320
7000	NX7-260	NX9-300	NX9-340	NX7-260	NX9-300	NX9-340	NX7-260	NX9-300	NX9-340
7250	NX9-290	NX9-320		NX9-290	NX9-320		NX9-290	NX9-320	
7500	NX9-300	NX9-330		NX9-300	NX9-330		NX9-300	NX9-330	
7750	NX9-330	NX9-360		NX9-330	NX9-360		NX9-330	NX9-360	
8000	NX9-340			NX9-340			NX9-340		

Note: max single panel length = 16 m

CLT Pre-Analysis Tables continued

Roof Pre-Analysis Tables

The recommended spans of the panels within the tables have been prepared on the basis of the following acceptance criteria. The designer may wish to increase or decrease some of these criteria based on the project demands.

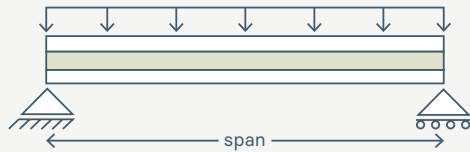
Live load taken as 0.25kPa

Permanent Load (Dead Load, including superimposed dead load), G	Span/250
Imposed Load (Live Load), Q	Span/300
Long Term, $j2 \times (G)$	The lesser of, Span/300 and 25 mm

Note that these tables are for ambient conditions, i.e. normal structural conditions, and a separate assessment needs to be undertaken for the fire load structural adequacy for the project's FRL period.

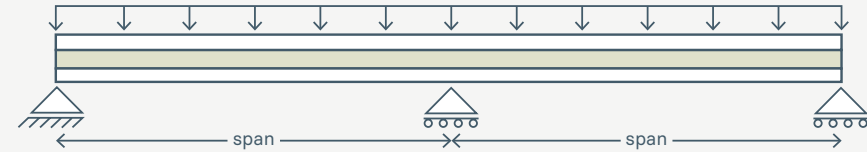


Roof – Pre-Analysis Tables



Single span

Super Imposed Dead Load (kPa)	0.0	0.5	1.0	2.0
Span (mm)				
2500	NX3-90	NX3-90	NX3-90	NX3-90
2750	NX3-90	NX3-90	NX3-90	NX3-90
3000	NX3-90	NX3-90	NX3-90	NX3-100
3250	NX3-90	NX3-90	NX3-90	NX3-100
3500	NX3-90	NX3-90	NX3-90	NX3-110
3750	NX3-90	NX3-90	NX3-100	NX3-120
4000	NX3-90	NX3-90	NX3-110	NX5-130
4250	NX3-90	NX3-100	NX3-110	NX5-140
4500	NX3-90	NX3-110	NX3-120	NX5-150
4750	NX3-90	NX3-110	NX5-130	NX5-160
5000	NX3-100	NX3-120	NX5-140	NX5-160
5250	NX3-100	NX5-130	NX5-150	NX5-180
5500	NX3-110	NX5-140	NX5-160	NX5-190
5750	NX3-110	NX5-140	NX5-160	NX5-200
6000	NX3-120	NX5-150	NX5-180	NX7-220
6250	NX5-130	NX5-160	NX5-180	NX7-220
6500	NX5-140	NX5-170	NX5-200	NX7-230
6750	NX5-140	NX5-180	NX7-210	NX7-240
7000	NX5-150	NX5-190	NX7-220	NX7-250
7250	NX5-160	NX5-200	NX7-220	NX7-260
7500	NX5-170	NX7-220	NX7-230	NX7-280
7750	NX5-180	NX7-220	NX7-250	NX9-290
8000	NX5-200	NX7-240	NX7-260	NX9-310



Double span

Super Imposed Dead Load (kPa)	0.0	0.5	1.0	2.0
Span (mm)				
2500	NX3-90	NX3-90	NX3-90	NX3-90
2750	NX3-90	NX3-90	NX3-90	NX3-90
3000	NX3-90	NX3-90	NX3-90	NX3-90
3250	NX3-90	NX3-90	NX3-90	NX3-90
3500	NX3-90	NX3-90	NX3-90	NX3-90
3750	NX3-90	NX3-90	NX3-90	NX3-90
4000	NX3-90	NX3-90	NX3-90	NX3-100
4250	NX3-90	NX3-90	NX3-90	NX3-100
4500	NX3-90	NX3-90	NX3-90	NX3-110
4750	NX3-90	NX3-90	NX3-100	NX3-110
5000	NX3-90	NX3-90	NX3-100	NX3-120
5250	NX3-90	NX3-90	NX3-100	NX5-130
5500	NX3-90	NX3-100	NX3-110	NX5-140
5750	NX3-90	NX3-100	NX3-120	NX5-140
6000	NX3-90	NX3-110	NX3-120	NX5-150
6250	NX3-90	NX3-110	NX5-130	NX5-150
6500	NX3-90	NX3-120	NX5-140	NX5-160
6750	NX3-100	NX3-120	NX5-140	NX5-170
7000	NX3-100	NX5-130	NX5-150	NX5-180
7250	NX3-100	NX5-140	NX5-150	NX5-180
7500	NX3-110	NX5-140	NX5-160	NX5-190
7750	NX3-120	NX5-150	NX5-170	NX5-200
8000	NX3-120	NX5-160	NX5-180	NX7-220

Note: max single panel length = 16 m

CLT Pre-Analysis Tables continued

Wall Pre-Analysis Table

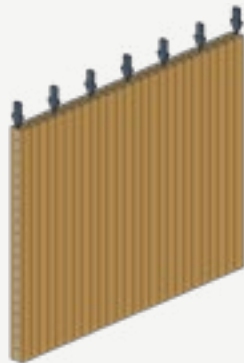
Axial Capacity of Walls

CLT walls offer high load-bearing capacity in comparison to conventional timber framing.

Preliminary Sizing Table Limitations

The indicative capacity published in the tables considers the orientation of outer lamellas to be vertical. In some instances, it may be suitable to specify a horizontally spanning wall (i.e. transverse layup), where it may be required to act as a deep beam. The values in these tables do not cover this application.

- Eccentricity is taken as thickness/15 and Height/500 for out of vertical
- P-delta (moment induced from second order effects)
- Slenderness ratio < 150
- Capacities based on Composite Sections in AS 1720.1 and FPInnovations Guide 2019 - Chapter 3.6 - Resistance of CLT Walls
- Load duration factor, $k_1 = 0.8$



Note that these tables are for ambient conditions (i.e. normal in-service structural conditions) and a separate assessment needs to be undertaken for the fire load structural adequacy for the project's FRL period. Also, the capacities do not consider the transmission of axial forces through the floor elements and consideration of concentrated loads adjacent to openings of building discontinuities that must be applied as part of the structural assessment.

Load (kN/m)	Wall Panel Height					
	3000	3250	3500	4000	4500	5000
50	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-100
100	NX3-90	NX3-90	NX3-90	NX3-100	NX3-110	NX3-120
150	NX3-100	NX3-100	NX3-110	NX3-120	NX5-130	NX5-140
200	NX3-110	NX3-110	NX3-120	NX5-140	NX5-140	NX5-160
250	NX3-120	NX5-130	NX5-140	NX5-140	NX5-160	NX5-170
300	NX5-130	NX5-140	NX5-140	NX5-160	NX5-170	NX5-190
350	NX5-140	NX5-140	NX5-150	NX5-170	NX5-190	NX7-210
400	NX5-140	NX5-150	NX5-160	NX5-180	NX5-200	NX7-220
450	NX5-150	NX5-160	NX5-170	NX5-200	NX7-220	NX7-230
500	NX5-160	NX5-170	NX5-180	NX7-210	NX7-220	NX7-240
550	NX5-170	NX5-180	NX5-200	NX7-220	NX7-230	NX7-250
600	NX5-180	NX5-200	NX7-210	NX7-220	NX7-240	NX7-260
650	NX5-200	NX7-210	NX7-220	NX7-230	NX7-250	NX7-280
700	NX7-220	NX7-220	NX7-220	NX7-240	NX7-260	NX9-290
750	NX7-220	NX7-220	NX7-230	NX7-250	NX7-280	NX9-300
800	NX7-220	NX7-230	NX7-240	NX7-260	NX9-290	NX9-310
850	NX7-230	NX7-240	NX7-250	NX7-280	NX9-300	NX9-320
900	NX7-240	NX7-250	NX7-260	NX9-290	NX9-310	
950	NX7-250	NX7-250	NX7-280	NX9-300	NX9-320	
1000	NX7-250	NX7-280	NX9-290	NX9-300		

Note: outer layers vertical

Glue Laminated Timber Range

Technical Specification – Glue Laminated Timber (GLT)

Wood Species	Australian Grown Radiata Pine (Pinus Radiata) Kiln Dried Softwood
Mechanical Grading	In accordance with AS/NZS 1748.1 and AS/NZS 4490
Manufacturing Standard	AS/NZ 1328.1
Joint Group	JD5
Strength Group	SD6
Moisture Content ex. Factory	10-15%
Adhesive – Face	Melamine Formaldehyde
Adhesive – Finger Joints	Melamine Formaldehyde
Lamella Thickness	42 mm
Stock Lamella Widths*	65, 85, 115, 135 mm
Beam Depth	126 to 1200 mm
Maximum Length	12 m
Feedstock Perseverative Treatments	Untreated and preservative treated to Hazard Class 3 in accordance with AS/NZS 1604.1
Density	460 to 540 kg/m ³
Fire Hazard Properties – Group Number and Average specific extinction area¹	Group 3 < 10 m ² /kg
Fire Hazard Properties – Critical Radiant Flux and Smoke Development rate²	3.5 kW/m ² and 7 % – min
Spread of Flame and Smoke Development Index	Radiata pine: 8 and 3

*NeXTimber® GLT press has the capacity to laminate beams to a maximum dimension of 1000 mm x 300 mm. Section width greater than the stock lamella widths will be assessed on a project by project basis.

1. An Assessment of Solid Timber Wall and Ceiling Linings in Accordance with AS 5637:2015 Report Number: RIR 45980.10

2. Fire Hazard Properties of Timber Floor, Wall and Ceiling Linings in Accordance with the Requirements of NCC 2019 Volume One Report Number: RIR 41117.9

Mechanical Properties – Glue Laminated Timber (GLT)

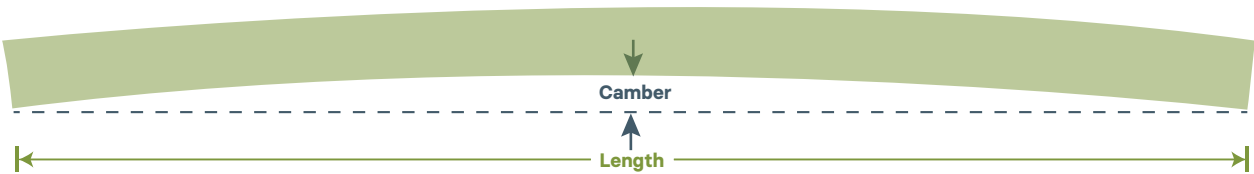
	Orientation	GL13
Modulus of Elasticity		13,300 MPa
Modulus of Rigidity		900 MPa
Bending Strength		33 MPa
Tensile Strength¹	Parallel	16 MPa
	Perpendicular	0.5 MPa
Shear Strength	Member	4.2 MPa
	At Joint	4.2 MPa
Compressive Strength	Parallel	26 MPa
Bearing Strength	Parallel	30 MPa
	Perpendicular	10 MPa

1. For tension members with a cross sectional dimension greater than 150 mm, the properties are determined by multiplying the value in the table by (150/d)^{0.167}

Glue Laminated Timber continued

Cambered Profiles

NeXTimber® GLT beams can be supplied either straight or with an Australian industry standard 600m radius to compensate for initial deflection and are designated with 'C' following the grade as per the above designation table. NeXTimber® cambered beams come labelled, indicating the top / outer radius of the camber. Unless specifically instructed, the beam should not be installed with the camber facing downwards.



Cambered Profiles Dimensions

Length (m)	3	3.5	4	4.5	5	5.5	6	6.5	7	7.5
Camber (mm)	1.9	2.6	3.3	4.2	5.2	6.3	7.5	8.8	10.2	11.7

Length (m)	8	8.5	9	9.5	10	10.5	11	11.5	12	12.5
Camber (mm)	13.3	15.1	16.9	18.8	20.8	23.0	25.2	27.6	30.0	32.6

Product Specifications

Product Specification

Type	Glue Laminated Timber (GLT)	Cross Laminated Timber (CLT)
Timber species	Australian Grown Plantation Radiata Pine	
Lamella Finished Width (mm)	65, 85, 115, 135	
Lamella Finished Thickness (mm)	42	20, 30, 40
Certifications	Structural Timber Certification Scheme (STCS), Forest Stewardship Council® chain of custody certified (FSC-C117015), Programme for the Endorsement of Forest Certification™ (PEFC)/ Responsible Wood (RW).	Structural Timber Certification Scheme (STCS), Forest Stewardship Council® chain of custody certified (FSC-C117015), Programme for the Endorsement of Forest Certification™ (PEFC)/ Responsible Wood (RW).
Surface Finish	Dressed / Planed with 45 degrees arris to edges (not ends).	Edge-glued with mechanically calibrated surface (minimum 60 grit sandpaper). Sanding direction is perpendicular to grain on crosswise layups.

Characteristics*	GLT Feedstock Limits	CLT Feedstock Limits
Checks	Permitted	Permitted
Knots	Edge knots: Permitted Central knots: Permitted Hole: Permitted	Edge knots: Permitted Central knots: Permitted Hole: Permitted
Pitch Pocket	Permitted	Permitted
Shake	Permitted	Permitted
Skips	Permitted	Permitted
Grain Slope	1 cm per 15 cm of length	1 cm per 15 cm of length
Split	50% part width	50% part width
Stain	Not graded	Not graded
Wane	15% Edge and Face	Occasionally visible
Pith	Not graded	Not graded
Moisture Content	9-12% average	9-12% average

*Explanatory Notes

The defect limits applied to feedstock for both CLT and GLT is that applied to Timberlink's standard Machine Graded Pine (MGP) grades with limited changes, namely:

- Tighter wane limits, particularly on GLT, so that cavities on the side faces of beams are mainly avoided.
- Tighter limits on moisture content, a critical parameter in glued engineered wood products, particularly for pressing in a Radio Frequency press.

Cracks and splits: Solid wood products may crack and form joints as they dry. This is natural and unavoidable. Cracking noises may also occur. It is recommended to maintain stable room humidity (nominally 50%) for exposed surfaces. Exposed surfaces may show undesirable contrast due to cracks.

It is recommended to Inform building owners and provide instructions to all involved parties during construction to manage moisture content.

Validity: The surface qualities mentioned are valid at the time of delivery. These qualities pertain only to the covering layer and do not extend to the narrow sides. Exposed surfaces may require a small amount of on-site reworking if finished surface is intended to remain exposed.

^ Refer to NeXTimber® technical notes for further information.
Visit www.nexttimber.com.au to access.

NeXTimber® CLT Stairs

CLT Stairs

NeXTimber® CLT stairs are manufactured from a solid billet of CLT which is then machined using a Computer Numerical Control (CNC) to form treads. NeXTimber® CLT stairs are supplied ready to install and include prefabricated landings. NeXTimber® CLT Stairs can be used in both commercial and residential applications and provide a sustainable, efficient and lightweight alternative to traditional building materials.

Important Note: In order to satisfy the requirements of the National Construction Code (NCC), the addition of no signs and slip resistant surfaces may be required. Applying a wearing surface to the stair is also recommended to increase the durability and avoid superficial damage.

Terminology



Preliminary Sizing Charts

The following tables are provided for guidance on throat thicknesses recommended for various load applications.

Vertical Span	1760	2020	2290	2550	2810	3070	3340	3600	3860
Horizontal Span	2000	2300	2600	2900	3200	3500	3800	4100	4400
Approx No. Steps	8	9	10	12	13	14	15	16	18
Inclined Span	2464	2834	3204	3573	3943	4313	4682	5052	5422
Live Load (kPa)	Superimposed Dead Load (kPa)	Throat Thickness							
2.0	0.00	NX3-90	NX3-90	NX3-90	NX3-90	NX3-90	NX3-120	NX3-120	NX3-120
	0.25	NX3-90	NX3-90	NX3-90	NX3-90	NX3-120	NX3-120	NX3-120	NX5-150
	0.50	NX3-90	NX3-90	NX3-90	NX3-90	NX3-120	NX3-120	NX5-150	NX5-200
4.0	0.00	NX3-90	NX3-90	NX3-120	NX3-120	NX3-120	NX3-120	NX5-150	NX5-200
	0.25	NX3-90	NX3-90	NX3-120	NX3-120	NX3-120	NX3-120	NX5-150	NX5-200
	0.50	NX3-90	NX3-90	NX3-120	NX3-120	NX3-120	NX5-150	NX5-200	NX5-200

Stair	Throat	Tread	Total Thickness
NXS7-250	NX3-90	40-40-40-40	250
NXS7-280	NX3-120	40-40-40-40	280
NXS9-310	NX5-150	40-40-40-40	310
NXS9-360	NX5-200	40-40-40-40	360

Notes

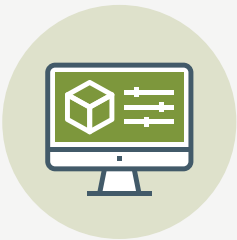
- Preliminary sizing in tables are intended to span from landing to landing. For preliminary transverse stair throat sizing, refer to Floor Pre-Analysis Tables.
- Sizes are suitable for preliminary sizing of stair throats - detailed design may yield different sizes
- Spans are based on a (R)ise of 180, a (G)oin of 250, and an incline of 35.8 degrees.
- Preliminary sizing does not take into consideration fire performance or requirements. A separate structural assessment is required.
- Sizes are based on the following performance criteria;
 - Short Term Dead Load – Span / 200
 - Short Term Live Load – Span / 250
 - Long Term ($j2 \times DL$) – Span / 400 or 25mm
- Vibration assessment: 8Hz or acceleration < 0.05ms⁻²
- All timber grades within the throat are to be MGP10 UNO.



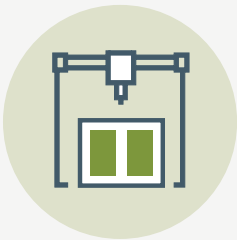
Post-Production Capabilities

Post-Production Capabilities

BIM & Design Integration



1. Model



2. Manufacture



3. Deliver

A fully integrated software solution will support project teams through concept and feasibility stages, costing, detailed design, manufacturing, logistics and installation.

An end-to-end platform allows customers to utilise Building Information Modelling (BIM), Computer Aided Design (CAD) and Computer Aided Manufacturing (CAM) to assist in delivering NeXTimber® projects efficiently and accurately.

Our in-house technical support team brings local and international mass timber project experience to provide an innovative, flexible, and supportive service for any type of project.

Post-Production Manufacturing – CLT

NeXTimber® CLT and GLT products are manufactured and fabricated within our facility in South Australia. Combining the efficiencies of digital technology and the accuracies of CNC capabilities, NeXTimber® products come pre-cut to finished dimensions, including all window and door openings, rebates, joints and lifting points.

Included in our supply offer is a standard level of CNC machining (i.e. large format panels, trimming and squaring of panels, door and window openings, panel to panel connections) and more extensive,

custom machining (i.e. concealed and intricate rebates for plates, service chases, cutting of small and intricate panels) can be provided at an additional cost and as required on a project-specific basis.

Machining Options

- Trimming and squaring of panels
- Panel to panel connections (Half lap, spline, butt joints)
- Inclined and angled cuts
- Rebates and slots for connecting plates and members
- Holes and penetrations, channels and rebates for electrical and plumbing services
- Pre-drilled holes and slots for fixings

Hundegger PBA Industry (Gantry CNC)

Tooling	5 axis circular saw
	5 axis milling machine
	3 axis milling machine
	5 axis chainsaw
	Drill sizes (various)
	End Mill (20 mm. 40 mm)
	Disk Mill (350 mm, 500 mm)
Dimensions	3.5 m wide x 16 m Long

Refer to the NeXTimber® website download library for further information on machining and DfMA

Connections & Lifting Solutions

Connections Supply

Mass timber connections are critical to the overall performance of the structure, and it is difficult for them to be assessed in isolation. As such, the project's structural engineer is responsible for the design, detailing and specification of mass timber elements and their connections.

There are several locally available mass timber fasteners and fixings suppliers distributed throughout Australia. They offer total fixing solutions (brackets, nails, screws and bolts) specifically developed for the mass timber industry – with well-published details and structural capacities.

NeXTimber® and fixing suppliers can provide guidance and technical advice on the use of specific details and hardware. However, the responsibility for the design solution remains the responsibility of the project's structural engineer. Refer to the NeXTimber® website download library for further information on project documentation

As part of our timber supply, NeXTimber® can also supply the specified structural connection hardware including off the shelf brackets and screws. Our team can prepare fixing takeoffs and supply fixings as part of the supply agreement. This service streamlines the process for our customers and ensures on-time supply as delivery of fixings and bracketry is scheduled in line with the timber manufacturing process.

Additionally, to support construction, NeXTimber® can supply an agreed lifting solution to ensure panels are ready to lift into place with minimal delay. Lifting devices can be pre-installed to facilitate safe and efficient construction once panels are delivered to the site.

Refer to the NeXTimber® website download library for further information.

Responsibility	Client	NeXTimber®
Building Design and Documentation	●	
Connection Details and Specification	●	
Structural Penalisation	●	
Engineering Design and Certification	●	
Refined Panelisation for Manufacture		●
Shop Drawings		●
Takeoffs		●
Supply of CLT, GLT & Connections		●

Connections & Lifting Solutions continued

Screw Connections

Self-tapping or self-drilling screws are generally preferred over bolts and brackets for cost, speed and ease of installation.

The capacities of fixings can be determined in accordance with the Australian timber engineering Standard AS 1720.1 or the Eurocode (EC5), which considers a multitude of failure mechanisms resulting in more efficient design. The characteristic capacity of fixings can be determined from the suppliers, or their European Technical Assessments (ETA) are available online.

Partially Vs. Fully Threaded Screws

Partially threaded screws are best used when elements need to be squeezed or pulled together. Typically, these are used for the majority of floor and wall connections.

Fully threaded screws generally provide higher tensile capacities than their partially threaded counterparts – however, as the thread extends along the full length of the shank, they won't clamp or squeeze elements together. Fully threaded screws are typically used for reinforcement of the timber or joint, increasing bearing resistance or where high tensile capacities are required.

Floor joints will need to resist floor diaphragm stresses and transfer lateral loads to the buildings bracing elements. Diaphragm irregularities may create a concentration of stresses and require specific details in localised areas and where high tensile capacities are required. In these instances, fully threaded screws may be used in combination once the partially threaded screws are installed.

It is critical to ensure that the minimum end and edge distance requirements for fixings into all timber elements (CLT & GLT) are met, particularly when connecting to steel elements which may be manufactured to different tolerances to timber products. We recommend that designers refer to the design standards (AS 1720.1 or EC5) or the supplier's ETAs for edge and end distance guidance, as this can differ from supplier to supplier for specific screw types and diameters.

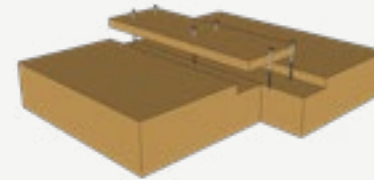
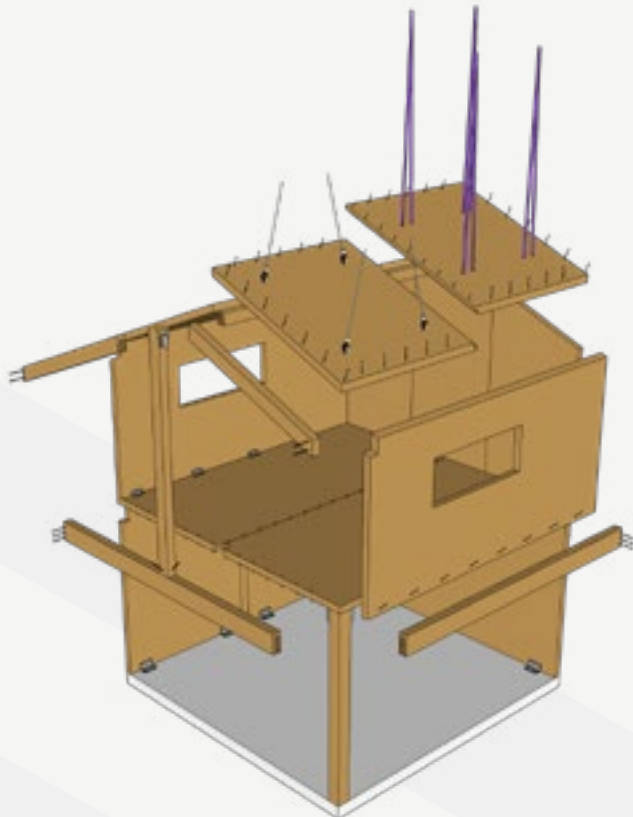


Connections & Lifting Solutions continued

Typical CLT and GLT Connections

NeXTimber® technical advisory services can be engaged early in the design process to support project engineers and building design teams in utilising the most effective connections. NeXTimber® works with project teams to model the connections into 3D manufacturing models to develop shop drawings, quantity takeoffs and machining files. NeXTimber® can then supply the required 3rd party fixing systems with the mass timber elements to the site in the required sequence.

Examples of typical CLT and GLT connections are as follows:



01 – Panel-Panel Spline Connection

Walls, floors and roofs



02 – Panel-Panel Half Lap Connection

Walls, floors and roofs



03 – Panel-Panel Connection at Ridge Beam



04 – Panel-Panel Connection at Ridge Beam



05 – Panel-Panel Connection

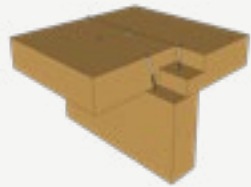
Walls and roofs

Connections & Lifting Solutions continued



06 – Panel-Panel Connection

Roof to wall, wall to wall



07 – Panel joint-Wall / Panel joint-Beam

Floors, roofs



08 – Wall-Floor / Floor-Wall Connection



09 – Panel-Wall Connection

Floor, roof



10 – Wall-Floor / Floor-Wall Connection / Wall-Slab



11 – Beam-Panel



12 – Beam-Panel



13 – Beam Connection



14 – Post Base Connection



Performance Requirements

Fire Resistance Levels (FRL) and Fire Resistance Ratings (FRR)

For NeXTimber® products, the fire resistance capacity has been independently assessed based on a series of fire tests conducted in different configurations - with and without fire grade plasterboard.

This assessment has been carried out by a registered testing authority in accordance with AS 1530.4. and is the basis of compliance evidence for NeXTimber® CLT and GLT.

Australian Standard AS 1720.4 has a one-dimensional char rate that can be used to calculate and determine fire resistance levels for solid timber and glue laminated timber elements. However, this standard scope does not presently include CLT or melamine bonded GLT and therefore is not suitable for NeXTimber® CLT or GLT.

NeXTimber® uses Melamine Formaldehyde (MF) adhesives (commonly referred to as Melamine) for all joints and laminations, including finger joints, edge, and face laminations which are cured under

pressure using an energy source. Melamine adhesives are common structural adhesives used in the manufacturing of timber products including CLT due to their structural performance and resistance to high temperatures.

One of the key benefits of melamine adhesives is that they maintain thermal structural stability past the combustion point of the timber. This means that the NeXTimber® laminated products, manufactured using melamine adhesives maintain their glue line integrity and prevents premature heat-induced char fall-off when exposed to fire.

Maintaining glue line integrity and minimising char fall-off are both critical to enhancing the fire performance of CLT panels, preserving the insulating properties of the char layer and delaying the fire's progression by protecting the underlying layers of wood, allowing the NeXTimber® panels to retain more of their load-bearing capacity.

Full scale fire tests conducted by Warringtonfire demonstrated that NeXTimber® products manufactured using melamine adhesives maintain glue line integrity and achieve a constant linear char rate when

subjected to high temperatures. This resulted in a predictable and high performing product with regard to fire performance as well as negating the need for the evaluation of complex, stepwise char models.

Details of fire resistant properties of NeXTimber® CLT and GLT products are covered in detail within the NeXTimber® Fire Performance Guide and Warrington Fire RIR for NeXTimber® products.



Acoustics

The acoustic separation between building compartments is a critical requirement of multi-residential construction, and the careful detailing of systems is required to meet the performance levels of the building code or expectations of the building occupants. As mass timber structures are relatively lightweight, they perform differently to more heavy masonry or concrete structures, as each material reacts differently to the frequency range that noise has. Therefore, strategies used for masonry or concrete structures may not be as successful when used in mass timber structures.

Airborne Noise

Airborne noise is defined as sound waves that move through the air. These sound waves can be in the form of speech, television, music. The acoustic performance of building elements can be improved by the introduction of additional dense materials such as dense flooring coverings, screeds, acoustic plasterboard or by separation or isolation.

Impact Noise

Impact noise is described as a structure-borne sound that has originated from impact. This structure-borne sound can be in the form of foot traffic or through dropping or moving objects. To reduce the effects of impact sound transmission, the introduction of materials that will absorb or reduce or deaden the effects of the impact will improve the acoustic performance. This reduction is commonly achieved through the use of resilient or absorbent materials, (i.e. carpet and acoustic underlays).

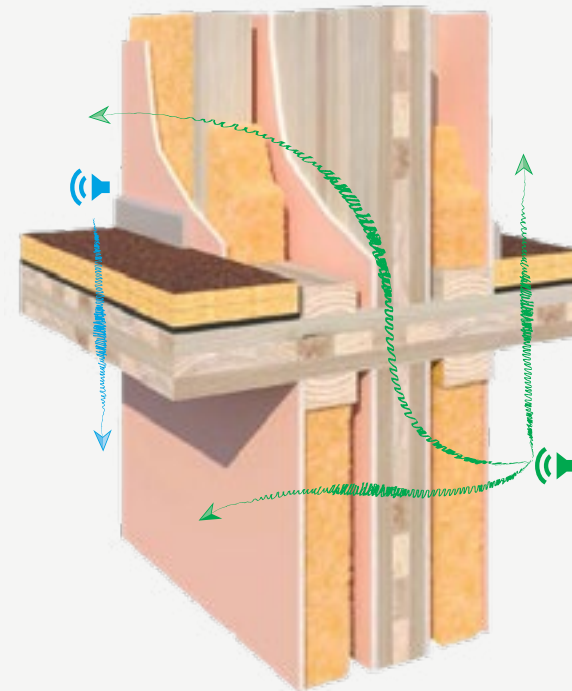
Flanking Sound

Flanking is the indirect transmission of sound waves or vibration through a structure. Flanking transmission paths can occur through mechanical exhausts, windows or openings or via indirect paths such as interconnected structural elements. Effective methods to mitigate flanking sound transmission include acoustic build-up on wall and floor assemblies, resilient separated ceilings and acoustically separated walls.

Flanking paths and overall acoustic performance need to be considered, particularly when considering exposed floors and walls in a building that may have sensitive acoustic performance requirements.

Tested and code compliant systems are available within Woodsolutions Guide 44 in addition to the tested and assessed solutions published with NeXTimber® Acoustic Performance Guide. Refer to the NeXTimber® website download library for further information.

Sound Transmission Paths



Moisture Control and Building Envelope

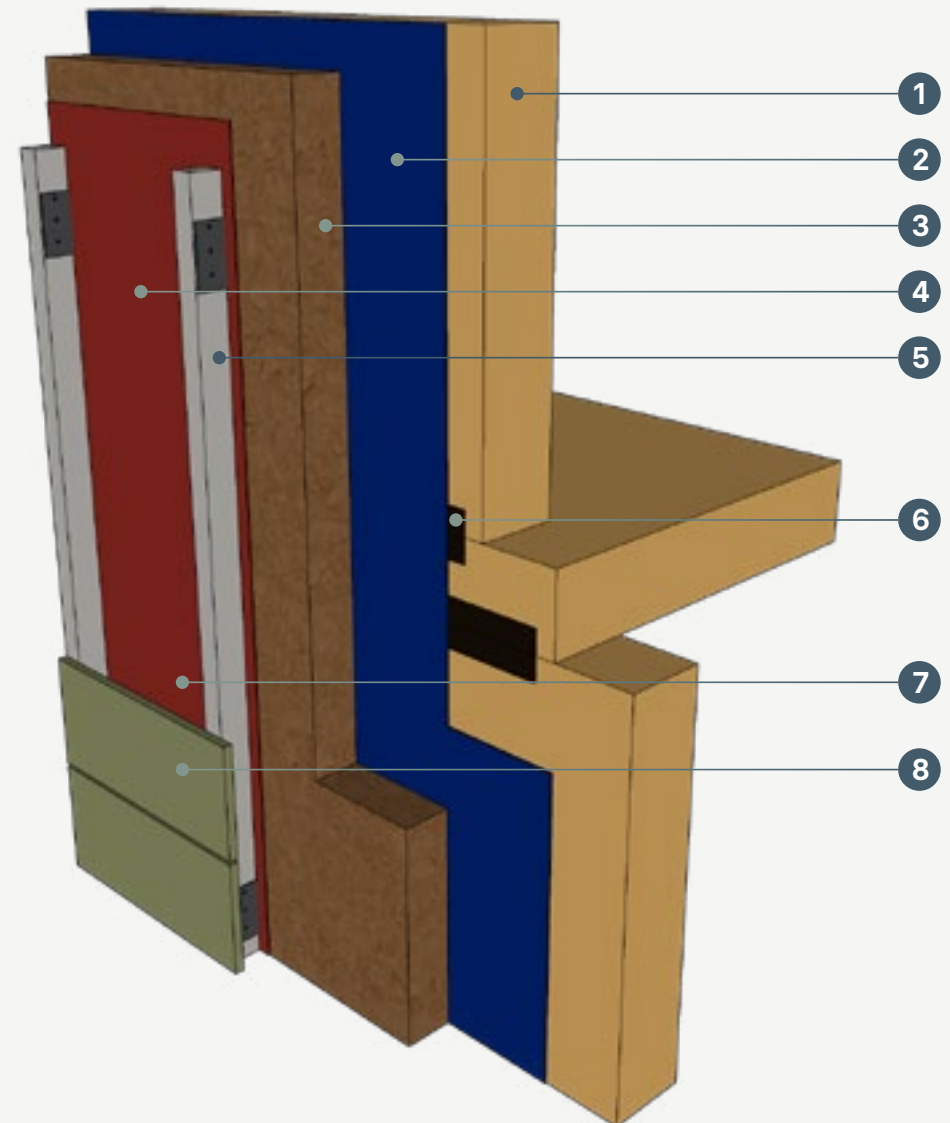
Moisture Content and Fungal Decay

One of the primary considerations for achieving long-term durability of timber buildings is mitigating the risk of developing fungal decay. Fungal decay can develop when the moisture content of timber products remains above 20% for an extended period. In well-designed and constructed timber structures, the moisture content typically ranges between 10% and 14% during service, significantly below the level where fungal growth and decay becomes a risk.

CLT construction follows the same durability principles as conventional timber framed structures. However, due to the thickness and mass of CLT this can introduce additional factors to consider when it comes to moisture exposure and control. For instance, if CLT becomes saturated, its large cross-sections can result in

significantly slower drying times compared to conventional timber frames meaning extra care around detailing is needed.

Irrespective of if the NeXTimber® CLT is treated, it should not be used in external above-ground or weather-exposed applications. NeXTimber® CLT manufactured using H3-treated feedstock may be used where the designers or customers determine it necessary due to either building code requirements or in higher-risk applications where an added level of protection is determined to be necessary (i.e. wet areas such as bathrooms and laundries). Use of H3-treated NeXTimber CLT is not a substitute for robust waterproofing practices and always consult your waterproofing specialists in these applications to achieve your desired durability outcomes.



- | | |
|-----------------------------|------------------------------------|
| 1 NeXTimber CLT | 5 Battens and fixings to substrate |
| 2 Vapour permeable Membrane | 6 Joint tape |
| 3 Insulation | 7 Drained and ventilated cavity |
| 4 Membrane | 8 Cladding |

Note: Refer to architect and suppliers for specification of building envelope suitable for your project requirements and climate. This image is indicative only and does not imply that all of these items will always be necessary, or that other items may be used.

Moisture Control and Building Envelope continued

Building Envelope

The building envelope plays an integral part in achieving a durable timber structure and performs four key functions:

- Moisture control
- Temperature / thermal control
- Air flow control
- Structural integrity (i.e. protection against airborne debris)

From a thermal comfort perspective (and to achieve the required 'R value' of the building envelope), additional insulation is generally required. This serves 2 key purposes:

- Increase the thermal properties of the building envelope assemblies
- Manages the risk of increasing the moisture content within the CLT to a level where fungal growth becomes an issue when designed, detailed and installed appropriately

Good design practices suggest that CLT external walls and roofs should always be designed as 'warm' construction. This means that insulation is placed on the outside face of panels, keeping the CLT within the building's thermal envelope. This should aid in keeping

the CLT inside a warm and dry environment allowing the timber to at a reasonably stable moisture content - below the point at which fungal growth occurs.

Building Envelope Ventilation and Breathability

Care is required in selecting the type and the placement of the additional insulation as it can affect the formation of condensation. Foil-backed or closed-cell insulation can be problematic in certain climatic zones if condensation forms within the envelope and cannot freely drain away or escape from the face of the CLT.

Issues can start to occur if timber becomes wet and its ability to dry is impeded by the use of improper insulation products and/or impermeable vapour control barriers or if moisture becomes trapped and cannot escape. In these circumstances, this can lead to increased moisture content in the timber, creating an environment favourable to the potential development of fungal growth and decay. This is especially the case if panels are subjected to ongoing wetting events (through poor

detailing and/or incorrect specification of insulating materials) either during construction or in service. Should the timber get temporarily wet (i.e. during construction), it is generally not a significant issue provided the water can freely drain away, and that the timber has the ability to naturally dry. This may include delaying the applications of finishes or preventing increases in moisture content by applying temporary moisture protection membranes prior to being exposed to water.

To achieve long-term durability objectives, the design, detailing and construction of the building envelope should contain a combination of drainage and ventilation in addition to proper insulation selection and application of membranes.

For these reasons, consultation and advice from building envelope specialists should be sought to ensure the proposed NeXTimber® products and build-ups are suitable for the intended region and climatic condition.

It starts with a seed from nature, with soil, water and sunshine added.

It's what a better tomorrow will be built on, built out of, built with.

It's what the craftsmen, the draftsmen, the architect, the engineer – all those who shape tomorrow – should lean on.

If it can be renewable and store carbon, it should, and it can.

And if it can all be done quicker, quieter, smarter, and safer it should.

And it can.

You see, if we treat our buildings as living breathing things, then the living breathing things that inhabit them, will have a future to look forward to.

Everything we do today matters, for tomorrow will be where we all reside.

However you use NeXTimber®, you won't just be building a building.

You'll be creating a better future.

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Version 3
April 2026

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CLT and GLT illustrations
used throughout are
artist impressions

